

IEOR 4404: Simulation

Lecture 9: Simulation Output Analysis

Point Estimator and Confidence Interval

- Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. sample from a population with an unknown parameter, say μ
- If $\mu = E[X_i]$, then a *point estimator* for μ (i.e., a single number that summarizes our best guess) is $\bar{X}$
- However, a point estimate cannot tell us the reliability of our best guess

Goal: estimate an unknown μ .

Point estimator: best guess of μ derived from data.

$\bar{X}$ is a good point estimator of μ because of law of large numbers if n is large.

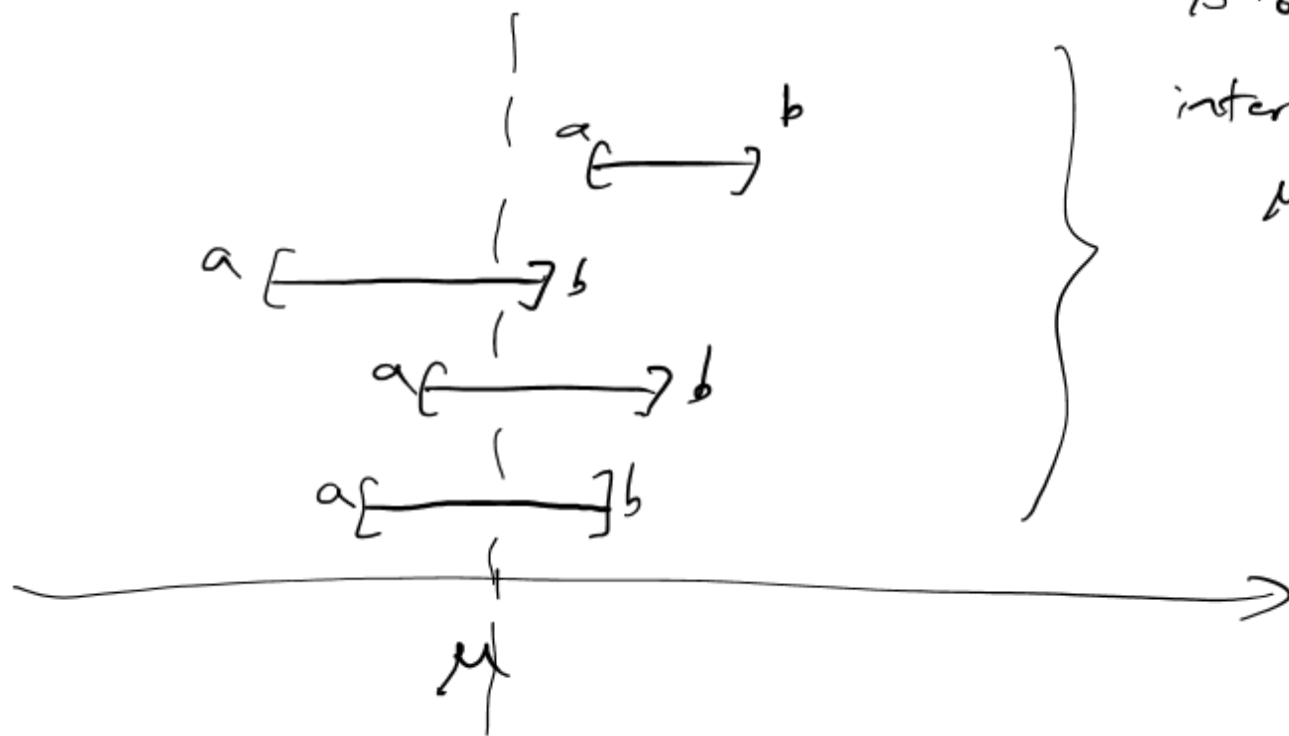
Confidence Interval

- Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. sample from a population with an unknown parameter, say μ
- To quantify error, we need an interval estimate, or confidence interval: Find two numbers, say a and b , from the simulation replications or data, such that the interval $[a, b]$ contains μ with high confidence, e.g.,

$$P(a \leq \mu \leq b) = 1 - \alpha$$

- This is called the $(1 - \alpha)$ -level *confidence interval*
- Typically $1 - \alpha = 95\%, 90\%, 99\%$ etc.

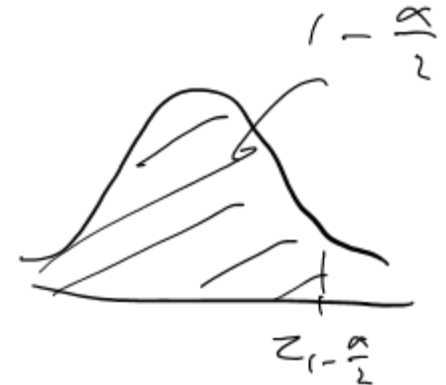
$$P(a \leq \mu \leq b) = 95\%$$



Confidence Interval: Known σ

- Suppose $X_1, X_2, \dots, X_n$ is a sample from a population with unknown mean μ and known variance σ^2
- The $(1 - \alpha)$ confidence interval is

$$\left[\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right]$$



where $z_{1-\frac{\alpha}{2}}$ is called the z-critical value = $\left(1 - \frac{\alpha}{2}\right)$ -quantile of $N(0,1)$, i.e.,

$$P(Z \leq z_{1-\frac{\alpha}{2}}) = 1 - \frac{\alpha}{2}$$

E.g.,

- for 95% confidence interval, $\alpha = 5\%$, and we use $z_{0.975} = 1.96$
- For 90% confidence interval, $\alpha = 10\%$ and we use $z_{0.95} = 1.645$

Do we need X_i to follow a normal distribution?

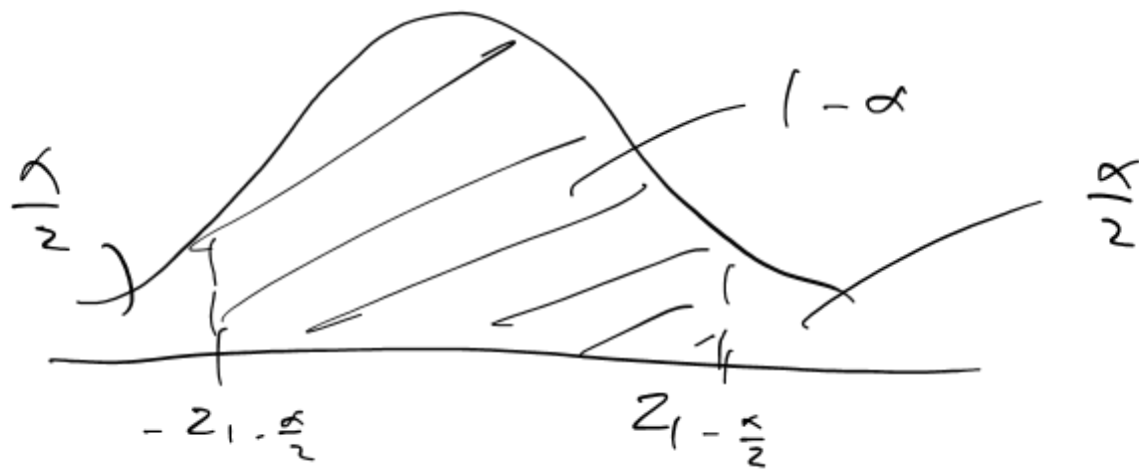
$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

This approximately holds when n is large, i.e. $n \geq 30$,
by CLT.

This holds exactly when X_i are normally
distributed.

$$\frac{\bar{X} - \mu}{\sigma} \sim N(0, 1) = Z$$

$$P(-z_{1-\frac{\alpha}{2}} \leq Z \leq z_{1-\frac{\alpha}{2}}) \approx 1 - \alpha$$



$$P\left(-z_{1-\frac{\alpha}{2}} \leq \frac{\bar{x} - \mu}{\sigma} \leq z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(\bar{x} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

So $\left(\bar{x} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right)$ is a

$(1 - \alpha)$ confidence interval.

Confidence Interval: Known σ

Explanation:

- If X_i are normal or when CLT is applicable, we have

$$P\left(-x \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq x\right) \approx P(-x \leq Z \leq x)$$

where $Z \sim N(0,1)$

- Note that

$$P(-z_{0.975} \leq Z \leq z_{0.975}) = \Phi(z_{0.975}) - \Phi(-z_{0.975}) = 0.95$$

- So

$$P\left(-z_{0.975} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{0.975}\right) \approx 0.95$$

- Rearranging terms, this becomes

$$P\left(\bar{X} - z_{0.975} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{0.975} \frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

Confidence Interval: Known σ

When does this interval hold?

- When X_i are normal, the *coverage* holds exactly correct, for any n , i.e.,

$$P\left(\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

- When n is large enough ($n \geq 30$), CLT applies and the coverage holds approximately correct, i.e.,

$$P\left(\bar{X} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) \approx 1 - \alpha$$

Confidence Interval: Unknown σ

Define

- Sample variance $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- Sample standard deviation $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$

Confidence Interval: Unknown σ

- Suppose $X_1, X_2, \dots, X_n$ is a sample from a population with unknown mean μ and unknown variance σ^2

- If n is large, i.e., ≥ 30 , then we use

$$\left[\bar{X} - z_{1-\frac{\alpha}{2}} \frac{S}{\sqrt{n}}, \bar{X} + z_{1-\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \right]$$

as the $(1 - \alpha)$ confidence interval for μ

- Justified from the CLT and a plug-in of the sample variance to approximate the unknown variance

Output Analysis from Simulated Data

- Consider estimating $E[h(X)]$ where $h(\cdot)$ is some function and X is one (or more) random variable
- In practice, both $h(\cdot)$ and X can be very complicated
- We can estimate $E[h(X)]$ by repeatedly running simulation to obtain many random copies of $h(X)$, and
 - output the point estimate by taking the sample mean
 - output confidence interval
 - determine sample size

using our developed methods

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $E[\sin X]$, where $X \sim N(0,1)$, using 100 simulation runs

Generate $X_1, \dots, X_{100} \sim N(0,1)$

Calculate $Y_1 = \sin X_1, \dots, Y_{100} = \sin X_{100}$

Calculate $\bar{y} = \frac{Y_1 + \dots + Y_{100}}{100}$ as the point estimate

95% confidence interval:

$$\left[\bar{y} - 1.96 \times \frac{s}{\sqrt{100}}, \bar{y} + 1.96 \times \frac{s}{\sqrt{100}} \right]$$

where $s^2 = \frac{1}{100-1} \sum_{i=1}^{100} (Y_i - \bar{y})^2$

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $E[\sin X]$, where $X \sim N(0,1)$, using 100 simulation runs

Procedure:

- Generate 100 independent copies of $N(0,1)$ from computer (e.g., using built-in routine or acceptance-rejection), say $X_1, \dots, X_{100}$
- Evaluate $Y_1 = \sin X_1, \dots, Y_{100} = \sin X_{100}$
- Point estimate = $\bar{Y}$
- Also calculate sample standard deviation S of Y_i , and output 95% confidence interval

$$= \left[\bar{Y} - z_{0.975} \frac{S}{\sqrt{100}}, \bar{Y} + z_{0.975} \frac{S}{\sqrt{100}} \right]$$

Sample Size Determination

To guarantee:

- A (point) estimate is within an error ϵ from the truth with $1 - \alpha$ probability
- The half-width of the $1 - \alpha$ confidence interval is within ϵ
- The margin of error at the $1 - \alpha$ confidence level is within ϵ

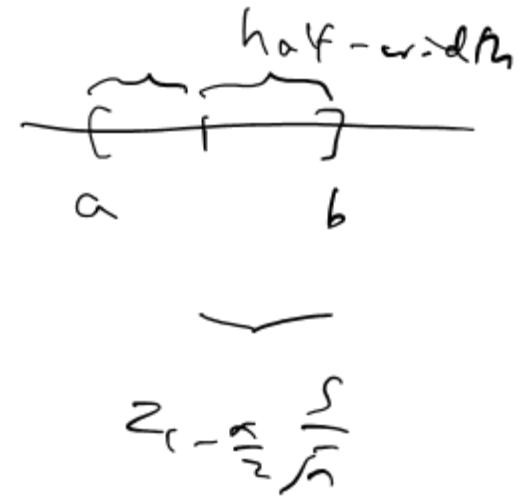
we need a sample size n such that

$$z_{1-\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \leq \epsilon$$

or that

$$n \geq \left(z_{1-\frac{\alpha}{2}} \frac{S}{\epsilon} \right)^2$$

We round up the required sample size n to an integer



Sample Size Determination

Notes:

- The sample variance S^2 needs to be estimated from a “pilot” set of data
- This procedure is valid only when the obtained n at the end is sufficiently large, i.e., ≥ 30

Sample Size Determination

To achieve a margin of error within ± 0.01 with 95% confidence in estimating $E[\sin X]$, where $X \sim N(0,1)$, how many simulation runs are needed?

$$n \geq \left(1.96 \times \frac{S}{0.01} \right)^2$$

where S^2 needs to be estimated from pilot runs.

Confidence Interval: Proportion

- Suppose the data are Bernoulli, i.e., each individual outcome is either a “success” (1) or “failure” (0).
- We are interested in estimating p , the proportion of success
- We can look at the sample proportion as the point estimate

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

where each data point X_i is either 0 or 1.

Confidence Interval: Proportion

- We also know that the variance of X_i is $p(1 - p)$, which can be approximated by $\hat{p}(1 - \hat{p})$
- Therefore, the $(1 - \alpha)$ confidence interval for p is

$$\left[\hat{p} - z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \hat{p} + z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \right]$$

- This is valid if n is a sufficiently large number, i.e., ≥ 30 so that a central limit theorem holds

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \sim N(0,1) \text{ approximately}$$

It gives almost the same answer.

• When n is small, the above interval fails because

X_i is not normal.

• Using previous intervals $\left[\bar{x} - z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{x} + z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right]$ also works.

Sample Size Determination: Proportion

To guarantee:

- A (point) estimate is within an error ϵ from the truth with $1 - \alpha$ probability
- The half-width of the $1 - \alpha$ confidence interval is within ϵ
- The margin of error at the $1 - \alpha$ confidence level is within ϵ

we need a sample size n such that

$$z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq \epsilon$$

$\hat{p} = \frac{1}{2}$ maximizes

or that

$$n \geq z_{1-\frac{\alpha}{2}}^2 \frac{\hat{p}(1-\hat{p})}{\epsilon^2}$$

$\hat{p}(1-\hat{p})$

We round up the required sample size n to an integer

Notes:

- The sample variance $\hat{p}$ is estimated from a “pilot” set of data; otherwise we simply take $\hat{p} = 1/2$ as this is the most conservative
- This procedure is valid only when the obtained n at the end is sufficiently large, i.e., ≥ 30

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $P(\sin X > 0.1)$, where $X \sim N(0,1)$, using 100 simulation runs

Generate $X_1, \dots, X_{100} \stackrel{iid}{\sim} N(0, 1)$

Calculate $Y_1 = I(\sin X_1 > 0.1) \dots Y_{100} = I(\sin X_{100} > 0.1)$

Calculate $\hat{p} = \frac{1}{100} \sum_{i=1}^{100} Y_i$ as point estimate

Output $\left[\hat{p} - 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{100}}, \hat{p} + 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{100}} \right]$

as 95% confidence interval

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $P(\sin X > 0.1)$, where $X \sim N(0,1)$, using 100 simulation runs

Procedure:

- Generate 100 independent copies of $N(0,1)$ from computer (e.g., using built-in routine or acceptance-rejection), say $X_1, \dots, X_{100}$
- Evaluate $Y_1 = I(\sin X_1 > 0.1), \dots, Y_{100} = I(\sin X_{100} > 0.1)$
- Point estimate = $\hat{p}$
- Output 95% confidence interval

$$= \left[\hat{p} - z_{0.975} \sqrt{\frac{\hat{p}(1-\hat{p})}{100}}, \hat{p} + z_{0.975} \sqrt{\frac{\hat{p}(1-\hat{p})}{100}} \right]$$

Output Analysis from Simulated Data

To achieve a margin of error within ± 0.01 with 95% confidence in estimating $P(\sin X > 0.1)$, where $X \sim N(0,1)$, how many simulation runs are needed?

$$n \geq 1.96^2 \times \frac{\hat{p}(1-\hat{p})}{0.01^2}$$

If we know $\hat{p}$, use the above
(pilot runs)

If we don't have pilot runs, plug in $\hat{p} = \frac{1}{2}$

and then round up.

Output Analysis from Simulated Data

To achieve a margin of error within ± 0.01 with 95% confidence in estimating $P(\sin X > 0.1)$, where $X \sim N(0,1)$, how many simulation runs are needed?

Answer:

- Since we don't have an estimate of p in advance, we take

$$n \geq \frac{z_{0.975}^2 \left(\frac{1}{2} \left(1 - \frac{1}{2} \right) \right)}{0.01^2} = 9604$$

- The minimum simulation size is 9604

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $E[(X_1^2 + X_2)^3]$, where $X_1, X_2 \stackrel{i.i.d.}{\sim} N(0,1)$, using 100 simulation runs

Generate $(x_1^{(1)}, x_2^{(1)})$, ..., $(x_1^{(100)}, x_2^{(100)})$

Calculate $Y_1 = (x_1^{(1)2} + x_2^{(1)})^3$, ...,

$Y_{100} = (x_1^{(100)2} + x_2^{(100)})^3$

Calculate $\bar{Y} = \frac{1}{100} \sum_{i=1}^{100} Y_i$

and $\left[\bar{Y} - 1.96 \times \frac{s}{\sqrt{100}}, \bar{Y} + 1.96 \times \frac{s}{\sqrt{100}} \right]$ where
 $s^2 = \frac{1}{100-1} \sum_{i=1}^{100} (Y_i - \bar{Y})^2$

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $E[(X_1^2 + X_2)^3]$, where $X_1, X_2 \stackrel{i.i.d.}{\sim} N(0,1)$, using 100 simulation runs

Procedure:

- Generate 100 independent copies of two i.i.d. $N(0,1)$ from computer (e.g., using built-in routine or acceptance-rejection), say $(X_1, \tilde{X}_1), \dots, (X_{100}, \tilde{X}_{100})$ where $(X_i, \tilde{X}_i)$ is a pair of i.i.d. $N(0,1)$
- Evaluate $Y_1 = (X_1^2 + \tilde{X}_1)^3, \dots, Y_{100} = (X_{100}^2 + \tilde{X}_{100})^3$
- Point estimate = $\bar{Y}$
- Also calculate sample standard deviation S of Y_i , and output 95% confidence interval

$$= \left[\bar{Y} - z_{0.975} \frac{S}{\sqrt{100}}, \bar{Y} + z_{0.975} \frac{S}{\sqrt{100}} \right]$$

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $P(X_1^3 + \dots + X_{50}^3 > 1)$, where $X_1, \dots, X_{50} \stackrel{i.i.d.}{\sim} N(0,1)$, using 100 simulation runs

Generate $(X_1^{(1)}, \dots, X_{50}^{(1)}), \dots, (X_1^{(100)}, \dots, X_{50}^{(100)})$

Calculate $Y_1 = I(X_1^{(1)3} + \dots + X_{50}^{(1)3} > 1)$

$\vdots$
 $Y_{100} = I(X_1^{(100)3} + \dots + X_{50}^{(100)3} > 1)$

Calculate $\hat{p} = \frac{1}{100} \sum_{i=1}^{100} Y_i$ and $\left(\hat{p} - 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{100}}, \hat{p} + 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{100}} \right)$

Output Analysis from Simulated Data

Give the point estimate and 95% confidence interval for $P(X_1^3 + \dots + X_{50}^3 > 1)$, where $X_1, \dots, X_{50} \stackrel{i.i.d.}{\sim} N(0,1)$, using 100 simulation runs

Procedure:

- Generate 100 independent copies of 50 i.i.d. $N(0,1)$ from computer (e.g., using built-in routine or acceptance-rejection), say $(X_1^{(1)}, \dots, X_1^{(50)}), \dots, (X_{100}^{(1)}, \dots, \tilde{X}_{100}^{(50)})$, where $(X_i^{(1)}, \dots, X_i^{(50)})$ are 50 i.i.d. $N(0,1)$
- Evaluate $Y_1 = I(X_1^{(1)3} + \dots + X_1^{(50)3} > 1), \dots, Y_{100} = I(X_{100}^{(1)3} + \dots + X_{100}^{(50)3} > 1)$
- Point estimate = $\hat{p}$
- Output 95% confidence interval

$$= \left[\hat{p} - z_{0.975} \sqrt{\frac{\hat{p}(1-\hat{p})}{100}}, \bar{Y} + z_{0.975} \sqrt{\frac{\hat{p}(1-\hat{p})}{100}} \right]$$

Estimating Quantities Other Than Mean

Quantities other than the mean, e.g., variance, covariance, correlation etc., can be estimated from simulation outputs or data via a natural sample counterpart

E.g., Given data $X_1, \dots, X_k$, estimate of their variance would be

$$\widehat{Var}(X) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \leftarrow \text{point estimate of } var(X)$$

This is justified by writing

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \underbrace{\left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2}_{E(X)} \approx E(X^2) - (E(X))^2 = var(X)$$

and using the law of large numbers

Estimating Quantities Other Than Mean

Given data $X_1, \dots, X_k$, estimate of the covariance between $g(X)$ and $h(X)$ would be

$$Cov(\widehat{g(X)}, h(X)) = \frac{1}{n} \sum_{i=1}^n (g(X_i) - \overline{g(X)})(h(X_i) - \overline{h(X)})$$

$\hat{= \frac{1}{n} \sum_{i=1}^n g(X_i)}$

and the correlation would be

$$Corr(\widehat{g(X)}, \widehat{h(X)}) = \frac{Cov(\widehat{g(X)}, \widehat{h(X)})}{\sqrt{Var(\widehat{g(X)}) Var(\widehat{h(X)})}}$$

$\hat{= \frac{1}{n} \sum_{i=1}^n h(X_i)}$

They all work, i.e. $Cov(\widehat{g(X)}, \widehat{h(X)}) \approx Cov(g(X), h(X))$

and $Corr(\widehat{g(X)}, \widehat{h(X)}) \approx Corr(g(X), h(X))$

because of law of large numbers.

When n is large, we can use " n " or " $n-1$ " in the sample var/cov formulas.

This is because $n \approx n-1$ when n is large.

When n is small, and we are in a situation where we need t -interval (population is normal, unknown var, and we want to estimate μ), then we use " $n-1$ ".